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University Management System

Feature Documentation

Role-based functional scope for Admin, Employee/Teacher, and Student

Document purpose

Provide an in-depth, mail-ready inventory of the features to be retained, modernized, and developed in the University Management System. The scope combines the approved UMS prototype with capabilities observed in the existing Amrita Academic Management Suite.

VERSION 1.0

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Status: Proposed functional baseline for stakeholder review and phased implementation.

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1. Document Overview

Purpose

This document defines what the new UMS should provide to its three primary user groups: administrators, employees/teachers, and students. It is designed for academic and technical stakeholder review before backend implementation.

> Important status note
> The modern UMS currently exists as an approved, navigable frontend prototype using mock data. Features in this document are the target product scope. Production authentication, permissions, APIs, business rules, and database persistence remain to be implemented.

Sources used

- Approved Amrita-inspired UMS frontend prototype.
- Existing AUMS screenshots showing Academics, Evaluation, Exam Admin, Survey, Reports, Personal, Search, Library, OPAC, Help, and Exam Application.
- Academic ERP practices required for secure role-based operation.

Role terminology

- Admin: academic, examination, department, library, or system administrators. Exact access depends on assigned permissions.
- Employee/Teacher: faculty and eligible academic employees responsible for courses, attendance, evaluation, mentoring, and reports.
- Student: enrolled learners using academic self-service features.

2. Feature Principles

- Role-based access: every screen and action is controlled by backend permissions.
- Single source of truth: academic records are central, effective-dated, and historical.
- Approval where needed: grade changes, attendance corrections, and result publication use workflows.
- Auditability: sensitive actions record actor, change, time, and reason.
- Operational clarity: every page handles loading, empty, success, validation, and error states.
- Responsive and accessible: core workflows support desktop, tablet, mobile, and keyboard operation.
- Controlled modernization: useful AUMS functions are retained with clearer navigation and reporting.

3. Role and Permission Model

Capability	Admin	Employee/Teacher	Student
Institutional data	Permission-based broad scope	Assigned classes/courses	Own records
Master data	Create/edit/activate	Relevant read access	No
Attendance	Configure/monitor/approve	Mark assigned classes	View own
Marks and grades	Configure/moderate/publish	Enter assigned-course marks	Published own
Reports	Institutional/departmental	Assigned-course scope	Personal scope
Library	Configure/report	Search and transact	Search and transact
Audit/settings	Authorized admins	No	No

Permission groups should be more granular than the three broad roles. For example, an Exam Controller may publish results without editing student master data, while a Department Chair may approve grade changes only within the department.

4. Admin Features

4.1 Admin Dashboard

- Show total students, faculty, courses, departments, active batches, and current semester.
- Show attendance trends, pending approvals, mark-entry completion, examinations, timetable conflicts, notices, and recent activity.
- Provide quick actions for users, course offerings, announcements, exams, and reports.
- Support campus, department, programme, year, semester, and date filters.
- Display only widgets permitted for the administrator's scope.

4.2 Identity, Roles, and Access

- Create, activate, suspend, unlock, and archive accounts.
- Assign roles, campus and department scopes, and action permissions.
- Support institutional SSO or secure local authentication, password policy, and MFA readiness.
- Configure session timeout, recovery, and login controls.
- Review login history, failed attempts, active sessions, and permission changes.
- Any impersonation capability must be restricted, visible, and fully audited.

4.3 University and Academic Master Data

- Maintain campuses, schools, departments, programmes, regulations, academic years, semesters, sections, buildings, rooms, laboratories, and holidays.
- Maintain staff records: employee ID, department, designation, employment status, contact information, and teaching eligibility.
- Maintain student records: ID, programme, batch, semester, section, status, mentor, and contacts.
- Maintain courses and subjects: code, title, credits, category, prerequisites, syllabus version, and owner.
- Maintain batches, capacity, curriculum/stem charts, programme duration, and semester structure.
- Use inactive/archive states instead of deleting historical records.

4.4 Curriculum, Course Administration, and Registration

- Create and version programme curricula and stem charts.
- Define core, elective, laboratory, project, audit, and value-added course categories.
- Create offerings with capacity, section, faculty, room, and delivery mode.
- Allocate faculty and identify workload or schedule conflicts.
- Configure registration windows, prerequisites, credit limits, add/drop, arrear rules, and approvals.
- Register students individually or in bulk and track incomplete registrations.
- Manage withdrawal, transfer, repeat registration, and section change with history.
- Report registration status, course demand, capacity, and exceptions.

4. Admin Features - Academic Operations

4.5 Timetable and Academic Calendar

- Build weekly timetables by programme, batch, section, faculty, room, and laboratory.
- Detect faculty, room, batch, and time-slot conflicts before publication.
- Configure slots, breaks, holidays, substitutions, and special sessions.
- Publish admin, teacher, and student timetable views with print/download/calendar export.
- Maintain dates for registration, teaching, assessments, exams, results, and events.

4.6 Attendance Administration

- Define thresholds, eligible session types, lock dates, shortage rules, and grace periods.
- Monitor capture completion by campus, department, course, faculty, and date.
- Review missing attendance, abnormal patterns, cancellations, and late entries.
- Process correction requests with reason, evidence, approval, and audit history.
- Produce daily, monthly, course-wise, student-wise, shortage, and eligibility reports.
- Notify teachers about pending entry and students about shortage risk.

4.7 Evaluation Configuration and Mark Governance

- Create components such as quizzes, assignments, labs, mid-semester exams, end-semester exams, projects, and viva.
- Set weights, maximum/pass marks, absence and rounding rules, and final-grade contribution.
- Create exams and configure mark-entry windows and authorized evaluators.
- Configure class grade rules, boundaries, moderation, and grade-mark categories.
- Track missing, absent, withheld, invalid, and completed mark entries.
- Assign grades, apply approved grace marks, and calculate final outcomes.
- Restrict grade changes to reasoned approval workflows with old/new values.
- Lock finalized records; authorized reopening must be audited.
- Generate class-grade, classwise student, attendance, registration-tracking, and grade-category reports.

4. Admin Features - Examination and Reporting

4.8 Examination Administration

- Create exam sessions, schedules, course eligibility, venues, and exam patterns.
- Allocate halls, seating, invigilators, and duties; publish timetables and hall tickets.
- Manage exam applications, eligibility, fee status, and approval.
- Track mark/grade entry status across courses and departments.
- Produce student-wise arrear, course-wise arrear, arrear-count, and performance reports.
- Manage absent, malpractice, withheld, incomplete, revaluation, supplementary, and improvement cases.
- Approve and publish results with version, publication time, and authorized sign-off.

4.9 Survey and Feedback Administration

- Create course, faculty, institutional, and custom surveys.
- Configure audience, anonymity, dates, mandatory status, and reminders.
- Track participation without exposing anonymous response identities.
- Produce not-submitted-user reports and aggregated analysis.

4.10 Reports and Analytics

- Student: profile, enrolment, registration, attendance, performance, progression, and arrears.
- Staff: profile, allocation, workload, timetable, attendance and mark-entry completion.
- Attendance: daily, course, student, shortage, exception, and correction history.
- Course offerings: capacity, enrolment, allocation, schedule, cancellation, and exceptions.
- Marks/grades: components, final marks, distributions, moderation, grace marks, and changes.
- Export authorized reports to PDF, spreadsheet, or CSV with filters and generation metadata.

4.11 Announcements and Notifications

- Create public or targeted notices by campus, department, programme, course, batch, or user.
- Schedule publication/expiry and support attachments, acknowledgement, and priority.
- Trigger messages for registration, timetable, attendance, assignments, exams, results, library dues, and approvals.
- Track in-app delivery/read status and support future email, SMS, and push channels.

4. Admin Features - Library and System Control

4.12 Library and OPAC Administration

- Maintain books, editions, authors, publishers, subjects, copies, locations, and digital resources.
- Manage issue, return, renewal, reservation, fine, lost/damaged, and exception workflows.
- Configure borrowing limits and loan periods by user category.
- Provide circulation, overdue, inventory, utilization, and fine reports.
- Connect the UMS entry point to OPAC or the approved library platform.

4.13 System Settings, Audit, and Operations

- Configure branding, academic defaults, numbering, formats, templates, integrations, and feature flags.
- Review audit logs by user, module, action, record, date, and outcome.
- Monitor integration failures, background jobs, imports/exports, and notification delivery.
- Support controlled imports with validation, preview, errors, and rollback planning.
- Define retention, archive, backup, restore, and disaster-recovery procedures.

Key administrative control
Mark entry, grade assignment, grade change, grace marks, result publication, attendance correction, and sensitive exports must each have a named permission owner and a complete audit trail.

5. Employee/Teacher Features

5.1 Teacher Dashboard

- Show today's classes, assigned subjects, student count, pending attendance, mark deadlines, submissions awaiting review, notices, and upcoming exams.
- Provide quick links to rosters, attendance, materials, assignments, marks, timetable, and reports.
- Display only allocated courses, classes, and authorized data.

5.2 Profile and Personal Account

- View employment profile, department, designation, qualifications, contacts, and responsibilities.
- Upload/update photo and edit permitted account fields.
- Request correction of protected HR-managed information.
- Access My Profile, View Profile, Update Account, and security/session controls.

5.3 Academics, Curriculum, and Timetable

- View stem charts, syllabus, learning outcomes, credits, and prerequisites.
- View and print personal/course timetable, room details, and academic calendar.
- Receive substitution, room-change, cancellation, and timetable-change notices.
- View workload and report allocation or schedule issues.

5.4 My Courses and Student Rosters

- View assigned offerings, sections, co-faculty, enrolment, and course settings.
- Search student rosters with permitted academic and registration details.
- Track registration and identify students who have not completed registration.
- Access course communication, materials, attendance, evaluation, and reports.
- Export authorized class lists without exposing restricted student data.

5. Employee/Teacher Features - Teaching Workflows

5.5 Attendance Capture

- Open a scheduled session and mark present, absent, late, or excused.
- Support bulk marking, individual correction before lock, and optional remarks.
- Show session, room, roster, and previously recorded status.
- Save draft where permitted and clearly confirm final submission.
- Request post-lock correction with reason and evidence.
- View history, shortage lists, student percentages, and pending sessions.
- Prevent entry for unauthorized courses or invalid dates.

5.6 Assignments and Study Materials

- Create assignments with instructions, files, marks, rubric, release/deadline, and late policy.
- Edit/cancel before cut-offs and notify affected students.
- View submissions, non-submitters, late work, file history, and remarks.
- Evaluate using marks/rubrics, add feedback, return work, and publish scores.
- Upload notes, presentations, references, links, manuals, and recordings.
- Organize materials by course, unit, topic, week, or type and control visibility.

5.7 Evaluation, Marks, and Grades

- View configured components and exams for assigned courses.
- Create components/exams only when policy and permission allow.
- Enter marks in a validated grid or authorized spreadsheet import.
- Record absent, withheld, malpractice, incomplete, or not-applicable status.
- Validate range, missing entries, duplicates, and completion before submission.
- View entry status and submit completed components for review/locking.
- Apply grade rules or assign grades only when permitted.
- Request grade changes with old/new values, reason, evidence, and approval trail.
- Apply grace marks only through authorized policy.

5. Employee/Teacher Features - Academic Services

5.8 Examination Responsibilities

- View examination timetable, venue, invigilation duty, evaluation allocation, and deadlines.
- Confirm or report conflicts in examination duties.
- Access authorized mark-entry and grade-entry tasks.
- View course arrear and performance reports only where the role permits.
- Submit examination-related requests and track status.

5.9 Reports

- Class grade and classwise student reports.
- Attendance history, percentages, shortage, and pending-entry reports.
- Course offering, registration status, marks, grades, and grade-category details.
- Grace-mark information only when explicitly authorized.
- Print/export restricted to assigned courses and permitted student attributes.

5.10 Survey, Search, Library, and Help

- Participate in assigned surveys; view non-submission reports only when authorized.
- Search permitted students, courses, faculty, rooms, and academic records.
- Search OPAC, reserve resources, view loans, renew items, and review fines.
- Access help, process guides, support requests, and announcements.

6. Student Features

6.1 Student Dashboard

- Display programme/semester, attendance, today's classes, exams, assignments, announcements, results, and library alerts.
- Highlight registration, low attendance, missing submission, exam application, dues, or profile-verification actions.
- Provide quick access to all student academic services.

6.2 Profile and Personal Services

- View identity, programme, batch, semester, section, mentor, contacts, and academic status.
- Update permitted contacts/photo and request correction of protected data.
- View account activity, active sessions, communication preferences, and consent.
- Download permitted identity or enrolment documents.

6.3 Curriculum and Course Registration

- View curriculum/stem chart, completed credits, pending requirements, prerequisites, elective baskets, and current offerings.
- Register for eligible courses within dates and credit limits.
- See clear block reasons: prerequisite, capacity, clash, hold, or approval.
- Request add/drop, withdrawal, repeat, or section change where permitted.
- Track approval and download the confirmed registration summary.

6.4 Courses, Timetable, and Calendar

- View registered courses, faculty, syllabus, outcomes, credits, and resources.
- View personal timetable with room, delivery mode, faculty, and schedule changes.
- Print/download timetable and add published events to a calendar.
- View deadlines, holidays, examinations, and university events.

6. Student Features - Academic Self-Service

6.5 Attendance

- View overall and course-wise percentages, counts, threshold, and shortage risk.
- View session history with date, time, course, faculty, and status.
- Receive alerts as attendance approaches or falls below policy.
- Submit correction requests with reason/evidence where allowed.

6.6 Assignments and Study Materials

- View upcoming, active, submitted, late, graded, and missed assignments.
- Read instructions/rubrics and upload permitted files before deadline.
- Save draft, replace submission before lock, and receive a dated receipt.
- Track marks, feedback, rubric outcome, and resubmission permission.
- Browse/download materials by subject, unit, topic, and date.

6.7 Examinations, Results, and Academic Record

- View exam schedule, venue, instructions, eligibility, and hall ticket.
- Apply for regular, supplementary, improvement, or other permitted exams.
- Track application, approval, payment, and hall-ticket status.
- View published marks, grades, SGPA, CGPA, credits, result status, and arrears.
- Download grade cards and permitted transcripts with verification details.
- Apply for revaluation/scrutiny within policy dates and track outcomes.
- Receive notifications when results are published or revised.

6.8 Survey, Notifications, Library, and Support

- Complete assigned course, faculty, programme, or institutional surveys.
- View targeted notices, events, deadlines, and emergency messages.
- Search own courses, materials, notices, faculty contacts, and public information.
- Search OPAC, reserve books, renew eligible loans, and view due dates/fines.
- Access help articles, FAQs, process guides, and support-request status.

7. Shared End-to-End Workflows

7.1 Course Offering and Registration

1. Admin prepares curriculum, offering, capacity, faculty, and schedule.
2. System validates prerequisites, clashes, credit limits, and eligibility.
3. Student registers or Admin performs an authorized bulk registration.
4. Required approvals route to the designated academic authority.
5. Confirmed registration appears in teacher rosters and student timetable.
6. Add/drop/withdrawal remains effective-dated in history.

7.2 Attendance

1. Published timetable creates eligible sessions.
2. Teacher records and submits attendance for the assigned class.
3. System updates percentages and identifies shortage risk.
4. Student sees the published record.
5. Post-lock corrections require request, approval, and audit entry.

7.3 Evaluation and Result Publication

1. Authorized user configures components, exams, weights, and grade rules.
2. Teacher enters and validates marks.
3. System tracks completion and flags missing/invalid records.
4. Authorized reviewer moderates, approves, applies policy-based grace marks, and locks.
5. Exam authority finalizes grades and publishes results.
6. Student sees published results; later change creates an auditable version.

7.4 Assignment

1. Teacher releases an assignment to an eligible course group.
2. Student submits and receives a dated confirmation.
3. Teacher evaluates and publishes marks/feedback.
4. Any resubmission retains previous submission history.

8. Reports and Outputs

Report family	Typical users	Key outputs
Student	Admin; teacher; student-own	Profile, registration, attendance, marks, grades, credits, progression, arrears.
Staff	Admin; authority; employee-own	Profile, allocation, workload, timetable, entry completion.
Attendance	Admin; teacher; student-own	Daily, course/student summary, shortage, missing entry, corrections.
Course offering	Admin; teacher	Capacity, enrolment, faculty, rooms, conflicts, exceptions.
Marks/grades	Exam/admin; teacher; student-own	Components, totals, distribution, entry status, changes, moderation.
Arrears/performance	Exam/admin; authorized teacher; student-own	Student/course arrears, counts, performance analysis.
Grace marks	Authorized exam/admin	Rule, beneficiaries, before/after, approver, audit.
Survey	Authorized admin	Participation, non-submitters where allowed, aggregate analysis.
Library	Library admin; user-own	Circulation, overdue, inventory, loans, reservations, fines.
Audit/security	Authorized system admin	Logins, permissions, sensitive changes, exports, failures.

Every export should include report title, filters, generation time, generated-by user, authorized data scope, and page/file integrity information where appropriate.

9. Notifications and Communication Events

- Account activation, security events, and profile-verification requests.
- Registration opening, approval, rejection, add/drop, and timetable change.
- Pending attendance, shortage, and correction decisions.
- Assignment release, deadline, receipt, grading, and feedback.
- Exam schedule, hall ticket, eligibility, application, and venue change.
- Mark deadlines, result publication, revaluation, and grade change.
- Library due/overdue, reservation, renewal, and fine.
- Survey invitation/reminder and targeted announcements.

10. Non-Functional Requirements

10.1 Security and Privacy

- Backend authorization, secure sessions, HTTPS, explicit CORS, validation, rate limiting, and secret management.
- Students access own records; teachers access assigned scopes; admins access permitted domains.
- Protect personal, academic, examination, and audit data in storage and transit.
- Log access to sensitive reports and exports with configurable retention.

10.2 Usability and Accessibility

- Responsive layouts, keyboard operation, visible focus, labels, readable contrast, and scalable text.
- Consistent patterns for dashboards, tables, forms, filters, dialogs, and reports.
- Search, sorting, pagination, bulk actions, and exports for high-volume pages.

10.3 Performance and Reliability

- Paginated APIs, database indexes, caching for stable references, and background exports.
- Idempotent submissions where duplicate academic operations would be harmful.
- Structured logs, monitoring, health checks, backup verification, and recovery procedures.

11. Integration Scope

- Institutional identity/SSO and directory services.
- Existing library platform and OPAC.
- Email, SMS, and push providers.
- Approved payment gateway for examination or academic fees.
- Learning Management System and digital-content platforms.
- HR, admissions, finance, and alumni platforms through governed APIs/imports.
- Calendar feeds and document-verification services.

Integrations should use documented APIs, environment-specific credentials, retry policies, monitoring, and reconciliation. Uncontrolled direct database coupling should be avoided.

12. Implementation Priority

Phase 1 - Foundation and Core Records

- Authentication, SSO readiness, roles, permissions, audit, and profiles.
- University masters, staff, students, programmes, courses, batches, subjects, and curriculum.
- Calendar, course offerings, registration, timetable, notifications, search, and reporting framework.

Phase 2 - Teaching and Assessment

- Attendance capture, correction, shortage monitoring, and reports.
- Assignments, submissions, feedback, and materials.
- Evaluation components, mark entry, grade rules, locking, and completion tracking.

Phase 3 - Examination and Outcomes

- Exam schedules, applications, hall tickets, duties, and eligibility.
- Grade assignment, grace-mark governance, arrears, results, grade cards, revaluation, and reports.

Phase 4 - Institutional Scale

- Library/OPAC, survey, advanced analytics, and approved integrations.
- Accessibility, security, performance, backups, pilot, training, and production launch.

13. Acceptance Baseline

- University owner approves user stories and permission rules.
- Frontend/backend builds, migrations, and automated tests pass.
- Validation, loading, empty, success, failure, and retry behavior is complete.
- Sensitive operations create audit events.
- Responsive, keyboard, accessibility, and browser behavior is verified.
- Affected reports and notifications are updated.
- Smoke tests and user-acceptance steps are documented.
- No mock data, debug content, secrets, or unauthorized data exposure remains.

14. Existing AUMS Reference Alignment

The supplied screenshots confirm the following existing AUMS navigation and functions. They have been retained in the proposed scope and reorganized into modern role-based workflows.

Existing area	Observed functions
Academics	Stem Chart; Print Staff Timetable; View Staff Timetable.
Evaluation	Create Component; Create Exam; Mark Entry; Class Grade Rule; Assign Grade; Change Grade; Class Grade Reports; Classwise Student Reports; Attendance Reports; Track Registration of Student; Grade Mark Category Detail; Apply Grace Mark.
Exam Admin	Mark/Grade Entry Status; Student Wise Arrear Report; Course Wise Arrear Report; Number of Arrears Report; Performance Analysis Report.
Survey	Survey Not Submitted Users Report.
Reports	Student; Staff; Attendance; Course Offering; Mark; Grade; Grace Mark reports.
Personal	Upload Photo; Update Account; View Profile; My Profile.
Additional	Dashboard; Course Admin; Attendance; Search; Library; OPAC; Help; Exam Application.

Stakeholder review request
Confirm which legacy features are mandatory for the first production release, which require process changes, and which may be retired. Also confirm permission owners for mark entry, grade assignment/change, grace marks, result publication, attendance correction, and sensitive reports.

Recommended review questions

- Which role currently performs each sensitive academic action?
- Which actions require department, school, or exam-controller approval?
- Which reports are statutory, accreditation-related, or operationally mandatory?
- Which current workflows depend on spreadsheet import/export?
- Which external systems are authoritative for identity, HR, admissions, finance, and library data?
- What historic data must be migrated and how long must it remain accessible?

UMS Feature Documentation - functional baseline for review and phased delivery.